Aerohive/Extreme Networks IQ Engine - Complete CLI Reference

Device: AP5050D-WW | Software: IQ Engine 10.8r5

Generated: December 22, 2025 **Purpose:** Comprehensive CLI command reference including hidden commands

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Overview

This document provides a complete reference of all CLI commands available on Aerohive (now Extreme Networks) Access Points running IQ Engine 10.8r5. This includes:

- Standard configuration commands
- Monitoring and show commands
- Executive/action commands
- File save/load commands
- Hidden debugging commands (starting with `_`)

CLI Access

- **Default Username:** admin
- **Default Password:** (varies by deployment)
- **Access Method:** SSH, Telnet (if enabled), Console

Command Syntax

- Use `?` after any command to see available sub-commands and options
- Use `Tab` for command completion
- Use `no` before most commands to negate/remove configuration
- Commands are case-sensitive

Configuration Commands (Root Level)

These commands are used to configure various aspects of the AP. Enter these at the root CLI prompt (`AFC-Outdoor-AP1#`).

Network & Interface Configuration

`802.1x-mac-table`

Set parameters for the client MAC address table that is used to track the status of authenticated clients and those attempting authentication through 802.1X/EAP

`aaa`

Set parameters for AAA (authentication, authorization, accounting)

`access-console`

Set access console parameters

`alg`

Set ALG (Application Level Gateway) parameters

`interface`

Set interface parameters / Set Energy Efficient Ethernet

`ip`

Set IP parameters

`ipv6`

Set IPv6 parameters

`dns`

Set DNS (Domain Name System) parameters

`routing`

Set routing parameters

`route`

Set a MAC address route

`lldp`

Set LLDP (Link Layer Discovery Protocol) parameters

Wireless & Radio Configuration

`acsp`

Set parameters for ACSP (Advanced Channel Selection Protocol)

`radio`

Set radio profile parameters

`ssid`

Set SSID (Service Set Identifier) parameters

`client-mode`

Set wireless client parameters for the local device so that it can associate with an access point

`client-load-balance`

Client load balancing parameters

`band-steering`

Band steering parameters in the WLAN

`high-density`

Parameters for optimizing performance in a high-density WLAN

`roaming`

Set roaming parameter

`amrp`

Set AMRP (Advanced Mobility Routing Protocol) parameters

Security Configuration

`security`

Set the security parameters

`security-object`

Set parameters for a security object controlling network access through the SSIDs and Ethernet interfaces to which it is applied

`supplicant`

Set parameters for a supplicant object for HiveOS

`network-firewall`

Set a Layer 3 firewall policy (Max: 1024 rules per Aerohive device)

`wips-essentials`

WIPS Essentials feature

User & Authentication

`admin`

Set the administrator parameters

`user`

Add one user or change user parameters

`user-group`

Set user group parameters

`user-profile`

Set parameters for a user profile

`user-profile-policy`

Set the user profile mapping policy

`login`

Set parameters for the CLI login

Management & Monitoring

`hive`

Create a hive or set hive parameters

`hiveui`

Enable the NetConfig UI for defining network settings, configuring settings to connect to HiveManager, and uploading a new HiveOS image

`hivemanager`

HiveManager parameters

`hostname`

Set the hostname of the AP

`management`

Set management service parameters

`snmp`

Set SNMP (Simple Network Management Protocol) parameters

`logging`

Set logging parameters

`report`

Set the parameters for gathering traffic statistics and reporting them to HiveManager

`telegraf`

Set telegraf configuration parameters

QoS & Performance

`qos`

Set QoS (Quality of Service) parameters

`cac`

Set CAC (Call Admission Control) parameters for regulating the admission of new VoIP calls

`performance-sentinel`

Set performance sentinel parameters to moderate client throughput

`forwarding-engine`

Set parameters to shape the behavior of the forwarding engine

Application & Service Configuration

`application`

Set L7 related parameters

`service`

Set a custom service

`bonjour-gateway`

Set parameters for the device to act as a Bonjour Gateway, collecting, filtering, and sharing Bonjour services across subnets/VLANs

`hotspot`

Set hotspot parameters

`library-sip-policy`

Set a SIP (Standard Interchange Protocol) policy to apply a user profile, VLAN, and session length to library patrons accessing the wireless network

Client & Device Management

`client-monitor`

Set parameters for Client Monitor

`client-tracing`

Test client tracing

`device-group`

Set a device group containing various objects that the HiveAP can use to classify client devices (Max: 64 groups)

`device-location`

Set the device location

`domain-object`

Set parameters for a domain object that the HiveAP can use to assign a client that belongs to a matching device domain to a user profile (Max: 64 domain objects per HiveAP)

`mac-object`

Set parameters for an MAC object that the HiveAP can use to assign a client with a matching MAC address to a user profile (Max: 128 MAC objects per HiveAP)

`os-object`

Set parameters for an OS object that the HiveAP can use to assign a client running a matching OS to a user profile (Max: 64 OS objects per HiveAP)

`os-detection`

Set the OS (Operating System) detection parameters

`os-version`

Set the OS (operating system) version you want to detect in the DHCP packets

`mdm-object`

Set the MDM (mobile device management) object

`mobile-device-policy`

Set a policy that assigns a user profile to traffic from a client based on the originally assigned user profile or the MAC address, device domain, and OS of the user's client

Policy Configuration

`ip-policy`

Set IP policy parameters

`mac-policy`

Set MAC policy parameters

`mobility-policy`

Set parameters for a mobility policy

`mobility-threshold`

Set parameters for tunneling mobile user traffic

`filter`

Set packet capture filter parameters

VPN & Tunneling

`vpn`

Set parameters for VPN (virtual private network) tunneling

`capwap`

Set parameters for CAPWAP (Control and Provisioning of Wireless Access Points)

`gre-tunnel`

GRE (Generic Routing Encapsulation) tunnel parameters

`ssh-tunnel`

Set SSH (Secure Shell) tunnel parameters so that Extreme Technical Support can access the AP remotely

`radsec-proxy`

Set parameters for RadSec proxy servers which communicate with ID Manager over a secure TLS tunnel

Data Collection & Analytics

`data-collection`

Set parameters for collecting data about the types and capabilities of devices on the network and the types of applications and IP protocols they use

`location`

Set parameters for location tracking

`location-essentials`

Location Essentials feature

`presence`

Set presence profile parameters

`ftm-policy`

Set parameters for FTM (Fine Time Measurement) policy

Network Services

`ntp`

Set NTP (Network Time Protocol) parameters

`clock`

Set the internal clock

`dhcp`

DHCP parameters

`web-directory`

Create a web directory for the internal web server

`web-security-proxy`

Proxy HTTP and HTTPS traffic bound for the Internet to a web security server for filtering / Set web security proxy service

`webui`

Set the WebUI on the HiveAP for local hive administration

Advanced Features

`iot-esl-imagotag`

Set iot-esl-imagotag parameters

`sdr-profile`

Set SDR (Software Defined Radio) profile parameters

`teacher-view`

Set parameters for TeacherView, a tool for controlling student access to the network and monitoring their activity

`probe`

Set the probe parameters

`sflow`

Set sflow related parameters

System Configuration

`system`

Set system parameters

`boot-param`

Set parameters for the boot loader

`config`

Set parameters for the current configuration file, which is a flash file containing default and admin-defined settings

`console`

Set console parameters

`license`

Set license parameters

`reset-button`

Enable the reset button on the AP chassis to reset the AP config

`validate_server_cert`

Set the validation of the server certificate parameter

Scheduling

`schedule`

Set a schedule to control the application of user profiles and the availability of SSIDs

`time-object`

Set a time object

USB & Hardware

`usbmodem`

Set parameters of usbmodem

`usbport`

Set the USB port

`pse`

Set PSE (power sourcing equipment) power management parameters

Testing & Diagnostics

`capture`

Set packet capture parameters

`iperf`

Set parameters for Iperf2, a tool for testing and measuring network performance

`iperf3`

Set parameters for Iperf3, a tool for testing and measuring network performance

`track`

Set parameters to track the reachability of one or more devices on the network

`track-wan`

Set parameters to track the reachability of one or more devices through the WAN interface

`net-access`

Set parameters to measure the internet available and latency

`rtts`

Create RTTS (Real Time Trouble Shooting) session

Miscellaneous

`adsp-server`

Set ADSP server ip address

`designated-server`

Set parameters for a dynamic server

`history`

Set the capacity for command history

`vlan-group`

Set a VLAN group

`kddr`

Enable/disable the kddr report to HM

`debug`

Enable debug messages

`reboot`

Set the system to scheduled reboot state

`reset`

Return the configuration to its default settings or the files in a web directory to the default file set

`fabric-attach`

Fabric attach configuration

Control Commands

`exec`

Execute a command to initiate a task immediately

`show`

Show settings, parameters, or dynamically generated information

`save`

Save a configuration, HiveOS image, RADIUS database, or files

`load`

Load a configuration file

`clear`

Clear dynamic system information or remove all web directories

`no`

Remove a configuration or return to a default setting

`exit`

Exit from the current mode

`quit`

Quit CLI (Command Line Interface)

Show Commands

Display system information, status, and configuration. All commands start with `show`.

Authentication & Security

`show 802.1x-mac-table`

Show the MAC table used for 802.1X/EAP user authentication on an Ethernet interface

`show aaa`

Show parameters for AAA (authentication, authorization, accounting)

`show auth`

Show authentication parameters per interface

`show ca-cert`

Show all CA certificates that the local AP uses to validate the server certificate in cURL command, or specific certificate content and expiration/start time

`show security`

Show security parameters

`show security-object`

Show security object names and individual parameters

`show supplicant`

Show supplicant parameters

`show idm`

Show ID Manager information

`show icsa`

Show ICSA (International Computer Security Association) parameters

System Information

`show system`

Show system information

`show version`

Show information about the current and backup IQ Engine versions on the HiveAP and the HiveAP platform type

`show hw-info`

Show hardware information

`show clock`

Show the date, time of the internal clock

`show cpu`

Show the percentage of the CPU used in total, for system operations, and for processing user traffic

`show memory`

Show total, free, and used system memory statistics

`show boot-param`

Show boot parameter information

`show license`

Show license information

Configuration & Running State

`show running-config`

Show currently running configurations

`show config`

Show parameters for the current configuration file

`show cmds`

Show CLI (Command Line Interface) commands including ones derived from optional keywords

`show history`

Show command history

Network & Interfaces

`show interface`

Show interface and subinterface parameters

`show ip`

Show IP parameters

`show ipv6`

Show IPV6 parameters

`show dns`

Show DNS (Domain Name System) parameters

`show route`

Show route parameters

`show routing`

Show routing parameters

`show arp-cache`

Show arp cache table

`show ndp-cache`

Show ndp cache table

`show lldp`

Show LLDP (Link Layer Discovery Protocol) parameters

`show l3`

Show Layer 3 information

Wireless & Radio

`show radio`

Show radio profile parameters

`show ssid`

Show SSID (Service Set Identifier) profile names and individual profile parameters

`show client-mode`

Show client mode settings

`show acsp`

Show parameters for ACSP (Advanced Channel Selection Protocol)

`show roaming`

Show the roaming cache and neighbors

`show band-steering`

Show parameters for band steering in the WLAN

`show high-density`

Show parameters for optimizing performance in a high-density WLAN

`show client-load-balance`

Show parameters for client load balancing in the WLAN

Client & Station Information

`show station`

Show information about all stations or about the ongoing wireless activity of a specific station

`show client-monitor`

Show Client Monitor parameters

`show client-info-collection`

Show client information collection result

`show private-client-group`

Show parameters for private client groups

`show ppsk`

Show parameters of private-PSK

Management

`show admin`

Show admin parameters

`show hive`

Show hive parameters

`show hivemanager`

Show HiveManager parameters

`show hiveui`

Show settings of the NetConfig UI

`show management`

Show management service parameters

`show access-console`

Show access console status and parameters

`show console`

Show console parameter

`show snmp`

Show SNMP (Simple Network Management Protocol) parameters

`show logging`

Show logging information

`show report`

Show report parameters for traffic statistics

`show telegraf`

Show all telegraf configurations

Application & Services

`show application`

Show L7 information

`show application-essentials`

Show Application Essentials parameters

`show service`

Show details or counters about predefined and custom services

`show bonjour-gateway`

Show the settings and status of the Bonjour gateway

`show mdnsd`

Show MDNS daemon information

`show hotspot`

Show a list of hotspot profiles or information about a specific hotspot profile

QoS & Performance

`show qos`

Show QoS (Quality of Service) parameters

`show cac`

Show CAC (Call Admission Control) parameters

`show performance-sentinel`

Show performance sentinel parameters

`show forwarding-engine`

Show forwarding engine parameters

`show alg`

Show ALG (Application Level Gateway) information

`show amrp`

Show AMRP (Advanced Mobility Routing Protocol) parameters

VPN & Tunneling

`show vpn`

Show VPN information and VPN objects

`show capwap`

Show the settings and current status for CAPWAP

`show gre-tunnel`

Show GRE (Generic Routing Encapsulation) tunnel information

`show ssh-tunnel`

Show SSH (Secure Shell) tunnel parameters

`show radsec`

Show RadSec proxy information

Security & Firewall

`show network-firewall`

Show all rules in the Layer 3 firewall policy

`show wips-essentials`

Show WIPS Essentials parameters

`show web-security-proxy`

Show the web security proxy configuration

Users & Policies

`show user`

Show all user

`show user-group`

Show a user group parameters

`show user-profile`

Show parameters for a user profile

`show user-profile-policy`

Show parameters for a user profile mapping policy

`show user-profile-schedule`

Show the status of all user profile schedules

`show min-password-length`

Show the minimum password length

`show device-group`

Show all device group names or the settings of an individual device group

`show domain-object`

Show all domain object names or the device domains assigned to an individual domain object

`show mac-object`

Show all MAC object names or the parameters of an individual MAC object

`show os-object`

Show all OS object names or the operating systems assigned to an individual OS object

`show os-detection`

Display the OS (Operating System) detection configuration

`show mdm-object`

Show MDM object information

`show mobile-device-policy`

Show all mobile device policy names or the settings of an individual policy

Policies

`show ip-policy`

Show parameters for IP policy

`show mac-policy`

Show parameters for MAC policy

`show mobility-policy`

Show the parameters of all mobility policies

`show mobility-threshold`

Show the settings for tunneling mobile user traffic

`show library-sip-policy`

Display library SIP policy settings

Monitoring & Diagnostics

`show capture`

Show packet capture parameters

`show filter`

Show capture filter parameters

`show data-collection`

Show parameters for collecting data about the types and capabilities of devices on the network and their network usage

`show track`

Show IP tracking information

`show track-wan`

Show Wan interface IP tracking information

`show proxy`

Show proxy parameters

`show sflow`

Show sflow related parameters

Location & Presence

`show location`

Show parameters for location tracking

`show location-essentials`

Location Essentials information

`show presence`

Show presence profile parameters

`show ftm`

Show the parameters for FTM neighbors

`show ftm-policy`

Show the parameters of all FTM (Fine Time Measurement) policies or one specified policy

Scheduling

`show schedule`

Show information about previously defined schedules

`show schedule-in-detail`

Show detailed information about all previously defined schedules

`show ssid-schedule`

Show the status of all SSID schedules

`show time-zone`

Show time zone

NTP & Time

`show ntp`

Show NTP (Network Time Protocol) parameters

Web & Files

`show web-directory`

Show the files in a web directory

`show web-server-key`

Show web server key files information

`show icon-directory`

Show a list of icon directories or the contents of a single directory

IoT & Special Features

`show iot-esl-imagotag`

Show iot-esl-imagotag setting parameters

`show sdr-profile`

Show SDR (Software Defined Radio) profile parameters

`show teacher-view`

Show parameters for TeacherView

`show fabric-attach`

Show fabric attach information

USB & Hardware

`show usb-device`

Show information about the internal USB hub and any device connected to the USB port

`show usbmodem`

Show parameters of usbmodem

`show pse`

Show PSE (power sourcing equipment) power management parameters

Reboot & Status

`show reboot`

Show if the system is scheduled to reboot

`show reset-button`

Show the state of reset button to reset the AP

Advanced/Other

`show rtts`

Show RTTS information

`show video`

Show information about streaming video traffic

`show vlan-group`

Show the settings and status of VLAN groups

`show wan`

Show brd wan info

`show adsp-server`

Show information about ADSP Server

`show tech`

Show the output of many "show" commands that display all the important settings and runtime data

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Exec Commands

Execute immediate actions and tasks. All commands start with `exec` .

AAA & Authentication

`exec aaa`

Set parameters for AAA (authentication, authorization, accounting)

`exec auth`

Execute an auth module command

Testing & Monitoring

`exec antenna-alignment`

Set parameters for aligning a directional or sectional antenna connected to a radio in backhaul or dual mode with a specified peer

`exec client-monitor`

Monitor the activities of a client

`exec tcp-service`

Test TCP service status

`exec quick-track`

Quick Track

Mobile Device Management

`exec aerohive-check`

Check the enrollment status of a mobile device on the Aerohive MDM server

`exec airwatch-check`

Check the enrollment status of a mobile device on the AirWatch

`exec jss-check`

Check the enrollment status of a mobile device on the JSS (JAMF software server)

`exec mobile-device-manager`

Set the mobile device manager parameters

Data Collection

`exec data-collection`

Perform an action on the data collected about the types and capabilities of devices on the network

Packet Capture

`exec capture`

Initiate packet capturing

Network Tools

`exec interface`

Execute the command through a specific interface

`exec ssh-client`

Secure Shell client

Security & Access

`exec bypass-wan-hardening`

Disable WAN hardening to allow SSH, Telnet, and the remote sniffer tool to access the device over the WAN interface (Note: To restore WAN hardening, enter "no exec bypass-wan-hardening" or reboot)

System Operations

`exec delay-execute`

Delay the execution of commands for a period of time (Note: This does not affect "show" commands)

`exec active-alarms-resending`

Make device resend all active alarms to HiveManager

`exec ftm`

Trigger an FTM neighbor list update

User Groups

`exec user-group`

Execute a user-group command

Save Commands

Save configurations and files. All commands start with `save`.

Configuration Files

`save config`

Save a configuration from the HiveAP to a remote server, from a remote server to the HiveAP, or from DRAM to flash as the current or bootstrap config

Certificates & Keys

`save ca-cert`

CA-certificate for validating the server certificate in cURL command during HTTPS session

`save radius-server-key`

Save certificate files for the local Aerohive RADIUS server to use

`save server-files`

Save certificate and private key files used by the internal web and RADIUS servers and VPN from DRAM to flash memory for persistent storage after reboots (Note: For security reasons, these files are saved only in DRAM by default)

`save vpn`

Save a VPN certificate or private key file

`save web-server-key`

Save certificate files for the internal web server to use

`save supplicant`

Save files for wpa supplicant

Images & Software

`save image`

Save a IQ Engine image to the HiveAP

`save signature-file`

Remote image used for L7 application

Files & Content

`save capture`

Save a packet capture file stored locally to a remote server

`save dhcp-fingerprint`

Save a fingerprint file of DHCP options for client OS detection

`save icon-file`

Save an icon file for an OSU (Online Signup) registration server (Max file size: 64 KB, formats: JPG, PNG, BMP, GIF)

`save ssid`

Save a locally stored file to a remote server

`save web-page`

Save a file for use with the internal web server

`save track-action-cli`

Save an admin-defined CLI file from a remote server to the local device

User & Client Data

`save users`

Save private PSK (preshared key) configurations

`save private-client-group`

Save private client group information from an external server to the local device

IoT Devices

`save ble`

Select the Bluetooth low energy device

`save thread`

Select the Thread device

Hidden/Debug Commands

These commands are not displayed in standard help output and are intended for debugging and advanced troubleshooting.

`_debug-adspsensor`

Enable debug feature for adsp sensor

⚠️ **WARNING:** Hidden commands are typically used by support personnel and developers. Use with caution as they may affect system stability or produce unexpected results.

Common Utility Commands

Navigation & Help

- `?` - Display context-sensitive help
- `Tab` - Auto-complete commands and parameters
- `history` - View command history
- `exit` - Exit current mode or CLI
- `quit` - Quit the CLI session

Network Testing

`ping`

Perform a ping test to check network connectivity

Usage:

```
ping <ip-address|hostname>
```

`tracert` / `traceroute`

Perform a traceroute to trace the path packets take to a destination

Usage:

```
tracert <ip-address|hostname>
```

Configuration Management

`load`

Load a configuration file

`clear`

Clear dynamic system information or remove all web directories

Common uses:

- `clear arp-cache`
- `clear session`

`reset`

Return the configuration to its default settings or reset files in a web directory

⚠️ **WARNING:** This will erase custom configurations!

`reboot`

Reboot the Access Point

Usage:

```
reboot
```

Or schedule a reboot:

```
reboot <time>
```

Configuration Modifier

`no`

Negate or remove a configuration setting

Usage:

```
no <command>
```

Examples:

```
no interface mgt0 ip 192.168.1.1
no ssid "Guest-WiFi"
```

Tips and Best Practices

1. Always Save Your Configuration

After making changes, remember to save:

```
save config
```

2. Use Tab Completion

Press `Tab` to auto-complete commands and see available options.

3. Context-Sensitive Help

Type `?` at any point to see available commands or parameters:

```
AFC-Outdoor-AP1# ?
AFC-Outdoor-AP1# interface ?
AFC-Outdoor-AP1# interface mgt0 ?
```

4. Verify Before Applying

Use `show` commands to verify current configuration before making changes:

```
show running-config
show interface
show ssid
```

5. Use "show tech" for Troubleshooting

When working with support, use:

```
show tech
```

This displays comprehensive system information.

6. Configuration Backup

Regularly backup your configuration:

```
save config <remote-server-details>
```

7. Review Changes Before Saving

The system will prompt you to save changes when exiting. Review what changed:

```
show running-config
```

8. Hidden Commands Caution

Hidden commands (starting with `_`) should only be used when:

- Directed by support personnel
- You fully understand the implications
- In a lab/test environment

9. Command Abbreviation

The CLI accepts abbreviated commands as long as they're unambiguous:

```
sh ver          (instead of: show version)
conf t          (instead of: config)
```

10. Schedule Maintenance Windows

For production APs, schedule reboots and major changes during maintenance windows:

```
reboot <date-time>
```

Quick Reference Card

Most Common Commands

Task	Command
View system info	`show system`
View software version	`show version`
View running config	`show running-config`
View SSID list	`show ssid`
View connected clients	`show station`
View interface status	`show interface`
Ping test	`ping <ip>`
Trace route	`tracert <ip>`

Save configuration	`save config`
Reboot AP	`reboot`
View CPU usage	`show cpu`
View memory usage	`show memory`
Get all system info	`show tech`

Common Configuration Tasks

Task	Command
Set hostname	`hostname <name>`
Set IP address	`interface mgt0 ip <ip> netmask <mask>`
Create SSID	`ssid <name>`
Set admin password	`admin <username> password <password>`
Enable NTP	`ntp server <ip>`
Remove config	`no <command>`

Support & Resources

For additional help:

- **Extreme Networks Documentation**: https://www.extremenetworks.com/support/documentation/
- **Support Portal**: https://extremeportal.force.com/
- **Community Forums**: https://community.extremenetworks.com/

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For the most up-to-date information, always refer to the official Extreme Networks documentation for your specific software version.

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